

EXPERIMENTAL INVESTIGATIONS OF REGIMES OF BUBBLE FORMATION ON SUBMERGED ORIFICES UNDER CONSTANT FLOW CONDITION

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Regimes of bubble formation on a submerged orifice under constant flow conditions were investigated experimentally. The effects of orifice diameter, surface characteristics (contact angle and roughness) and surface tension on the regimes of bubble formation were studied for a wide range of gas flow rates. In particular, the transitions of period-1 to period-2 bubbling regime, with pairing or with coalescence at the orifice, and period-2 to chaotic bubbling were investigated in detail. An attempt is made to provide a generalized bubble formation regime map constructed using appropriate dimensionless numbers. The physical understanding of various bubbling regimes and the experimental data on the effects of various system parameters are expected to contribute to the development and validation of analytical and CFD models.

On a étudié de façon expérimentale les régimes de formation des bulles sur un orifice immergé dans des conditions d'écoulement constant. Les effets du diamètre d'orifice, des caractéristiques de surface (angle de contact et rugosité) et de la tension de surface sur les régimes de formation des bulles ont été étudiés pour une vaste gamme de débits de gaz. En particulier, les transitions de régime de bullage de période-1 à période-2, avec couplage ou avec coalescence à l'orifice, et du bullage de période-2 au bullage chaotique, ont été examinées en détail. On tente de fournir une carte généralisée des régimes de formation des bulles dressée à partir de nombres adimensionnels appropriés. La compréhension physique des divers régimes de bullage et les données expérimentales sur les effets de divers paramètres de système devraient contribuer au développement et à la validation de modèles analytiques et de CFD.

Keywords: bubble formation, dynamics, regimes of bubble formation, transition periods

INTRODUCTION

The bubble formation process is important to many gas-liquid operations that are found in chemical processing, biochemical operations, metallurgy and waste water treatment. In many of these applications, bubble formation is accompanied by transport processes (heat and mass transfer, for example, as in boiling and separation processes). In most of these processes, gas is introduced into the process equipment through *orifices* (as in sieve plate spargers) or *nozzles*. Very often, the performance of the underlying gas-liquid process or the contacting device depends on the size of the gas bubbles formed at the orifice or nozzle.

Owing to the importance of bubble formation processes in many practical applications, bubble formation has been an active area of multiphase flow research for the last 50 years. In most of the early investigations in this area, Davidson and co-workers studied the bubble formation processes extensively (Davidson and Amick, 1956; Davidson and Schueler, 1960a, b; Walters and Davidson, 1963). Depending on the mode of gas injection and liquid viscosity, the bubble formation process was

classified as: (a) constant flow; and (b) constant pressure with an inviscid or viscid bubble formation process (Davidson and Schueler, 1960a, b). Since then, several experimental investigations have been carried out to study bubble formation under the conditions of constant flow (von Krevelen and Hoftijzer, 1950; Davidson and Schueler, 1960a, b; Ramakrishnan et al., 1969; Terasaka and Tsuge, 1993) and constant pressure (Davidson and Schueler, 1960a, b; Satyanarayan et al., 1969). Fan and co-workers (Lin et al., 1998) studied the bubble flow characteristics, its instability in bubble columns. The effect of gas chamber volume (V_c) below the orifice plate on bubble formation was also studied (Kupferberg and Jamson, 1969; Park et al., 1977; Tsuge and Hibino, 1983; Terasaka and Tsuge, 1990).

Most of these investigations were carried out to elucidate the effect of orifice diameter (Davidson and Schueler, 1960a, b; Ramakrishnan et al., 1969; Satyanarayan et al., 1969; Tsuge and Hibino, 1983; Tsuge and Hibino, 1993), surface wetting

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characteristics (Liow and Gray, 1988), viscosity with either Newtonian or non-Newtonian liquids (Davidson and Schuele, 1960a; Ramakrishnan et al., 1969; Satyanarayan et al., 1969; Raebiger and Vogelpohl, 1982; Terasaka and Tsuge, 1990, 1993; Bork et al., 2005), surface tension (Davidson and Schuele, 1960b; Ramakrishnan et al., 1969; Satyanarayan et al., 1969; Terasaka and Tsuge, 1993), liquid density (Ramakrishnan et al., 1969) and physical properties of the gas phase (Tsuge and Hibino, 1983) on the volume of the bubble formed at the orifice at different gas flow rates. Kumar and Kuloor (1970) reviewed most of the early developments in this area. Kulkarni and Joshi (2005) have reviewed the recent work in the area of bubble formation and bubble rise behaviour. In recent years, efforts were also made to understand the non-linear characteristics of bubble formation processes or *chaos* in bubbling. Such non-linear characteristics were investigated using 3-D-attractors, correlation dimension and auto correlation functions (Tritton and Egdell, 1993; Mittoni et al., 1995; Nguyen et al., 1996; Tufaile and Sartorelli, 2000; Mosdorf and Shoji, 2003; Cieslinski and Mosdorf, 2005).

However, it can be seen from most of the above-mentioned literature that the dynamics of bubble formation, mainly the regimes of bubble formation, were not investigated in detail. It is unclear how the previously formed bubble influences the growth of the next bubble forming at the orifice and leads to different bubbling regimes. Further, the effect of various operating and design parameters on different bubbling regimes and on their transition characteristics is not understood. In most of the literature discussed above, except for the single period bubbling regime, the reported bubble volume represents the average bubble volume that results from double, triple or quadruple coalescence at the orifice (Ramakrishnan et al., 1969; Satyanarayan et al., 1969). In some of the reports, the average bubble formation frequency was also used to represent the bubble formation process (Cieslinski and Mosdorf, 2005).

Based on the visual observations, the regimes of bubble formation were qualitatively classified as period-1 (single) bubbling, period-2 bubbling, that is, bubbling with pairing (bubbles coalesce far above the orifice) and with coalescence (bubbles coalesce at the orifice itself before detachment), bubbling with triple or quadruple coalescence (period-3 or period-4 bubbling) and finally the chaotic bubbling (Leibson et al., 1956; McCann and Prince, 1971; Kyriakides et al., 1997; Tufaile and Sartorelli, 2000). In recent years, attempts were made to quantify regimes of bubble formation under constant flow conditions (Zhang and Shoji, 2001; Tufaile and Sartorelli, 2002; Buwa et al., 2005). However, these investigations were limited to one set of orifice diameter or physical properties of liquids. It is not understood how the above mentioned bubbling regimes and their transition occur with changes in the orifice diameter, surface characteristics and physical properties of the liquid phase.

Depending on the operating and design parameters [d_o , $U_o(Q_G)$, θ , σ , ρ , μ , P , V_c], the bubble formation process is governed by different forces. Of these, the lifting forces acting on a bubble are: buoyancy force $\left[F_B = \frac{\pi}{6} d_b^3 (\rho_l - \rho_g) g\right]$, pressure force $\left[F_P = \frac{\pi}{4} d_o^2 (\rho_g - \rho_l)\right]$, and force due to gas momentum $\left[F_M = \frac{\pi}{4} d_o^2 \rho_g U_o^2\right]$. F_M is small and can be ignored, except at high pressures and at high gas flow rates. The restraining forces which act in the opposite direction of the lifting forces are surface tension (or capillary) force $[F_S = \pi d_o \sigma]$, drag force $\left[F_D = \frac{\pi}{4} d_b^2 C_D \frac{\rho_l U_b^2}{2}\right]$ and inertial force $\left[F_I \approx \left(\frac{99}{32\pi} + \frac{9}{2\pi} \frac{\rho_g}{\rho_l}\right) \rho_l \frac{Q_G^2}{d_b}\right]$. At the time of bubble

detachment, lifting forces become equal to the restoring forces ($F_B + F_M = F_S + F_D + F_I$). By solving the above force balance, several different empirical and analytical models were developed to predict the bubble volume. At very low gas flow rates, the buoyancy force is balanced by the surface tension force (characterized by the Bond number, $Bo = \rho_l g d_o^2 / \sigma$, defined as the ratio of buoyancy force to surface tension force). This regime of bubble formation is known as static bubble formation. In this regime, the bubble volume is proportional to the orifice radius and surface tension and is independent of the gas flow rate (Gerlach et al., 2005). As the gas flow rate is increased, the surface tension force becomes negligible except at the early stages of bubble formation, and inertial force takes over the surface tension force. At higher gas flow rates (characterized by the Froude number, $Fr = U_o^2 / g d_o$, defined as the ratio of inertial force to gravitational force), the bubble formation frequency remains constant and the bubble volume is proportional to the gas flow rate and is nearly independent of surface tension.

Depending on the underlying assumptions and levels of complexity, simple one-stage models to more accurate two-stage models were developed to predict the bubble formation process. A few of these models (Davidson and Schuele, 1960b; Davidson and Harrison, 1963; Kumar and Kuloor, 1970; Gaddis and Vogelpohl, 1986; Jamialahmadi et al., 2001) are listed in Table 1. The predictions of these models for different gas flow rates are shown in Figure 1. It can be seen from Figure 1 that most of these models predict a monotonic increase in the bubble volume with increase in the gas flow rate. Most of the models developed to predict bubble formation are based on the assumption of single bubbling and do not consider different regimes of bubble formation (for example, see the review by Kumar and Kuloor (1970)). Such models can only be applied to predict the bubble volume at low gas flow rates at which single (periodic) bubble formation is observed. At higher gas flow rates, the previously formed (leading) bubble significantly influences the growth of the next forming bubble at the orifice, therefore affecting their

Table 1. Correlations developed to predict bubble volume

Reference	Correlation
Davidson and Schuele (1960b)	$V_b = 1.378 \frac{Q_G^{6/5}}{g^{3/5}}$
Davidson and Harrison (1963)	$V_b = 1.138 \frac{Q_G^{6/5}}{g^{3/5}}$
Kumar and Kuloor (1970)	$V_b = 0.976 \frac{Q_G^{6/5}}{g^{3/5}}$
Gaddis and Vogelpohl (1986)	$V_b = \frac{\pi}{6} \left[\left(\frac{6d_o\sigma}{\rho_l g} \right)^{4/3} + \left(\frac{8\mu_l Q_G}{\pi \rho_l g} \right) + \left(\frac{135Q_G^2}{4\pi^2 g} \right)^{4/5} \right]^{3/4}$
Jamialahmadi et al. (2001)	$V_b = \frac{\pi}{6} d_o^3 \left\{ \left[\frac{5}{Bo^{1.08}} \right] + \left[\frac{9.261 Fr^{0.36}}{Ga^{0.39}} \right] + \left[2.147 Fr^{0.51} \right] \right\}$
	$Bo = \frac{\rho_l g d_o^2}{\sigma}$
	$Fr = \frac{U_o^2}{g d_o}$
	$Ga = \frac{\rho_l^2 d_o^3 g}{\mu_l^2}$

formation periods. Under conditions of constant gas inflow, the bubble volume can be calculated directly from the formation period ($V_b = t_b \times Q_G$). It should be noted that the change in successive bubble formation periods can be significantly large (with maximum bubble volume 4–5 times larger than that of the minimum bubble volume, as will be shown later) after the period-1 bubbling regime (i.e., in period-2, period-3 and chaotic). Recently, Zhang and Shoji (2001) emphasized the need for such models and proposed a new non-linear bubble model to account for the non-linear interactions of the bubble forming at the orifice with the previously formed bubble. However, the applicability of such models has not yet been verified for a wide range of operating parameters (orifice diameter, surface wetting characteristics and physical properties of the liquid phase). Recently, detailed two-phase free surface flow models (based on volume of fluid (VOF) or Level set (LS) methods) were also applied for *a priori* simulation of such non-linear bubble formation processes (Buwa et al., 2005; Gerlach et al., 2006). However, progress in the development of these models is often limited by a lack of the experimental data to validate the computational models. In order to improve the present status of the modelling efforts, physical understanding of different bubbling regimes is important and need experimental data on the effect of different parameters on the regimes of bubble formation and their transition. The present work was undertaken to provide this information.

In many practical applications, the pressure fluctuations in the gas chamber below the orifice plate (for example the sieve plate spargers) can influence the bubble formation process. The liquid flow or the bulk liquid circulation can also influence the bubble formation processes. However, the objective of the present work is to understand the various regimes of bubble formation (based on the different bubble formation periods) and their transition, and to investigate the effect of various operating and design parameters on the regimes of bubble formation. Therefore, the scope of the present work is limited to the investigations of bubble formation on a single orifice under constant flow conditions. In this work, experimental investigations on the dynamics of bubble formation were carried out by varying orifice diameters (2, 4 and 6 mm) for a wide range of gas flow rates (75 to 3000 cm³/min). This corresponds to the high gas flow rate regime of Davidson and Schueler (1960a, b). The corresponding Bond and Froude numbers were in the range

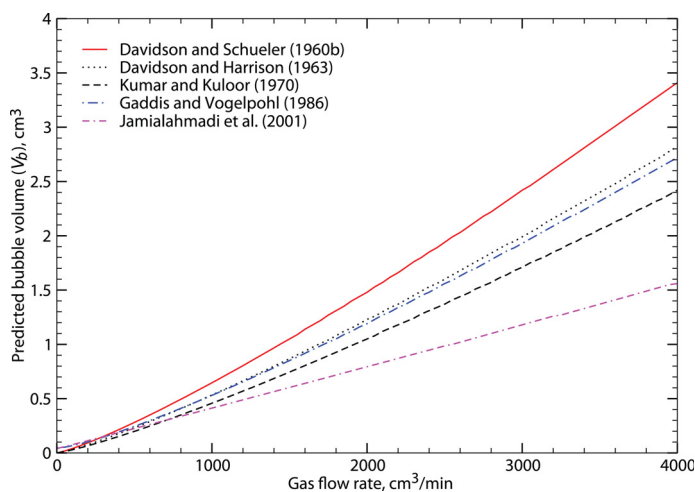


Figure 1. Predicted bubble volumes using different correlations available in the literature (see Table 1)

0.55–4.9 and 0.034–5738, respectively. All the investigations were carried out for a low-viscosity (air-water or air-2-propanol + water) system (the corresponding capillary numbers were in the range 0.002–0.46). Under the conditions of the present investigations, the bubble formation process is dominated by the inertial forces compared with the viscous forces. The effect of surface wetting characteristics was investigated by using different orifice plates, made of stainless-steel, Teflon and acrylic sheet. The effect of surface tension of the liquid was studied by varying the surface tension coefficient of liquids from 0.037 to 0.072 N/m. The effect of these parameters on the regimes of bubble formation and their transition was investigated experimentally. Finally, bubble formation regime maps constructed using the *Froude* and *Bond* numbers are provided.

EXPERIMENTAL WORK

The experimental set-up used to study the bubble formation dynamics is shown in Figure 2. A square cross-section (14 × 14 cm sides) glass column (70 cm in height) was used. In all experiments, distilled water was used as the liquid, unless mentioned otherwise, with the static liquid level maintained at around 10 cm from the bottom of the column with a corresponding liquid height (*H*) to column width (*W*) ratio less than 1. For *H/W* ratio less than 1, Delnoij et al. (1997) has shown that local re-circulatory flow is absent and bubble chain rise vertically upwards at the centre of the column without any meandering motion. Air bubbles were generated from an orifice plate submerged in the stagnant liquid as shown in Figure 2. A provision was made in the set-up to change the orifice plate so that the bubble formation on surfaces with different wetting characteristics could be studied. A 1/16-in stainless-steel capillary (inner diameter = 0.89 mm) was used to supply air from a mass flow controller to

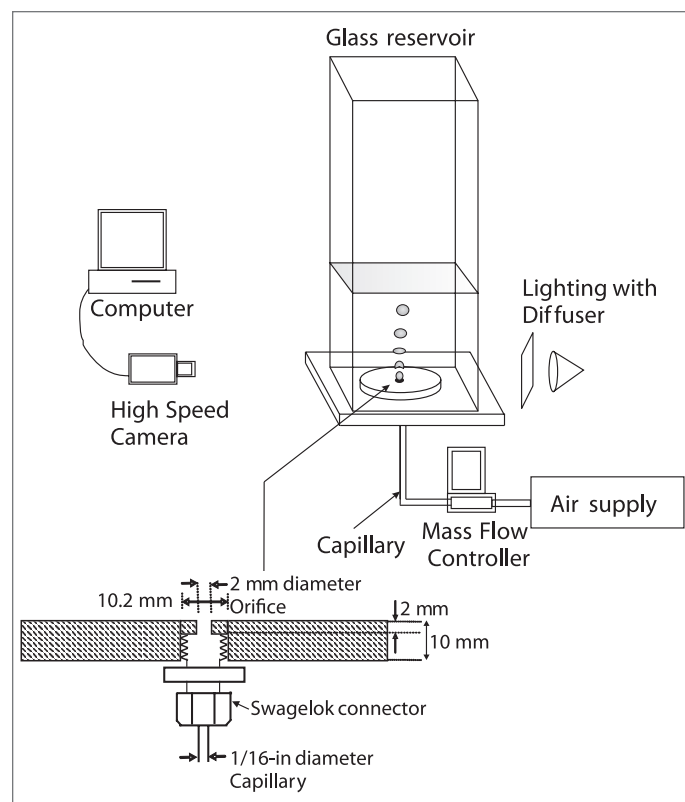


Figure 2. Schematic diagram of experimental set-up with sketch of the orifice assembly

the orifice. The capillary was connected to the orifice using swagelok connector (SPC SS-100-1-2RS) at the bottom of the orifice plate (see sketch of the orifice in Figure 2). The air volumetric flow rate was regulated with a mass flow controller (Bronk Horst Model F201C-FB-33V, Wagner M+R GmbH). A high-speed digital camera (Visario) was used to record the bubble formation process. The bubble formation process was recorded with a frame grabbing speed of 1000 frames/s. Acquired images were corrected using background subtraction and the resulting images were enhanced using image processing software (Scion Image, Matlab, Frame from AVI v3.1). Depending on the bubbling regime, the detachment times of successive bubbles were measured using the recorded frames with an accuracy of ± 1 ms.

Following Takahashi and Miyahara (1976), the length of the capillary was taken in such a way that constant flow conditions could be achieved. Takahashi and Miyahara showed that when $l_{\text{cap}}/(d_{\text{cap}})^4$ (where l_{cap} is the length of the capillary and d_{cap} is the inner diameter of the capillary) is greater than $1 \times 10^{12} \text{ m}^{-3}$, constant flow can be achieved. In the present work, a capillary with $d_{\text{cap}} = 0.89 \text{ mm}$ and $l_{\text{cap}} = 1 \text{ m}$ was used. Therefore, the ratio, $l_{\text{cap}}/(d_{\text{cap}})^4$ for the capillary used was $1.594 \times 10^{12} \text{ m}^{-3}$ and hence satisfied the above criteria. To ensure the constant flow conditions further, the increase in the individual bubble volume was calculated at a regular interval of time for more than three consecutive bubbles forming at the orifice and the results are shown in Figure 3. The bubble volume in Figure 3 was calculated using Guldin's second law (Harris and Stoecker, 1998). It can be seen that the volume of the bubble increases linearly with time and subsequent bubbles are formed without any time lag, indicating constant flow conditions at an airflow rate of about $75 \text{ cm}^3/\text{min}$. Further, the final bubble volumes calculated using Guldin's second law and that obtained from experiments (a product of measured bubble formation time and gas flow rate) agreed within $\pm 7\%$. Therefore, the gas flow rates for all the experiments performed in the present work were greater than $75 \text{ cm}^3/\text{min}$ at which the constant flow condition was ensured.

The effect of airflow rate on bubble formation was investigated by varying the airflow rate from 75 to $3000 \text{ cm}^3/\text{min}$. The effect of orifice diameter on the bubble formation process was investigated by considering orifices with diameters of 2 , 4 and 6 mm . The effect of surface characteristics (contact angle and

surface roughness) on the dynamics of bubble formation was studied using orifice plates made of stainless-steel, Teflon and acrylic sheet. The 3-phase, static contact angle of different orifice plates was measured using the OCA contact angle system (Model OCA20, Dataphysics Instruments GmbH) based on the principle of the sessile drop method and are listed in Table 2. The surface roughness was measured using a Taylor-Hobson (Sultronic) instrument and its average value at different locations on the orifice plate is also given in Table 2. The effect of surface tension on the formation process was also studied by taking different liquid systems with the addition of different amounts of 2-propanol to water. These results are discussed in the following section.

RESULTS AND DISCUSSION

Regimes of Bubble Formation: Effect of Gas Flow Rate

Initial experiments were performed to understand the various regimes of bubble formation that can be observed at different gas flow rates. At lower gas flow rates, bubbles were formed above the orifice at a regular interval of time and this regime of bubbling is known as the single bubbling regime (see Figure 4a). This regime of bubble formation is characterized by a single value of formation period (referred as the period-1 bubbling regime hereafter). The bubble shape is close to spherical form in all stages of the bubble formation in period-1 bubbling. It can also be seen that the previously formed bubble is sufficiently far from the bubble growing at the orifice so that the growth of the later bubble is not influenced by the wake behind the leading bubble.

With further increase in gas flow rate ($Q_G = 377 \text{ cm}^3/\text{min}$, see Figure 4b), the bubble formation period was decreased to 41 ms from 45 ms (which was observed at $Q_G = 94 \text{ cm}^3/\text{min}$). However, the rise velocity of the leading bubble was not increased proportionally. The growth of the next bubble at the orifice is therefore accelerated by the wake behind the leading bubble (bubble formation period = 39 ms , see Figure 4b). These two bubbles rise together until some distance above the orifice and eventually coalesce far above the orifice. The next bubble grows at the orifice without any influence of the wake behind the leading bubble and the period-2 bubbling repeats as shown in Figure 4b. This regime of bubbling is known as period-2 bubbling with pairing of bubbles. The onset of period-2 bubbling was considered when there was a minimum of 2 ms difference between a pair of successive bubble formation periods. With further increase in gas flow rate ($Q_G = 566 \text{ cm}^3/\text{min}$), the influence of the wake behind the leading bubble becomes so significant that the next bubble forming at the orifice elongates vertically and coalesces with the leading bubble during its formation. This regime is known as the period-2 bubbling regime with coalescence at the orifice. The formation period (38 ms) and volume of the second bubble are smaller than those of the first bubble (44 ms), see Figure 4c.

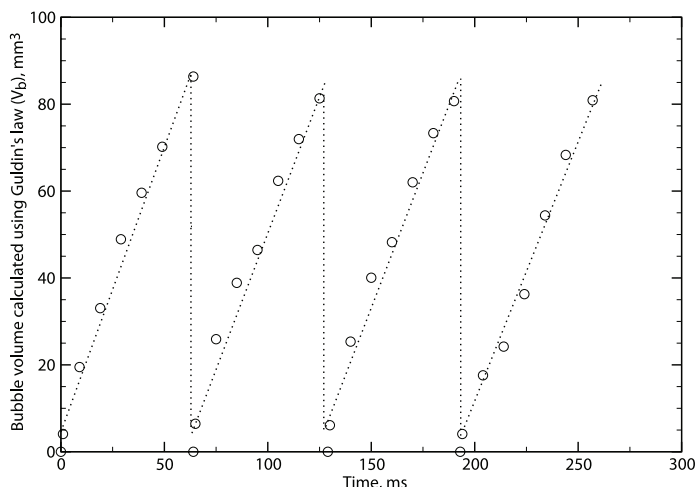


Figure 3. Constant flow condition: evolution of bubble volume with time ($d_o = 4 \text{ mm}$, $Q_G = 75 \text{ cm}^3/\text{min}$, stainless-steel orifice, liquid: water) [lines are drawn to provide a guide to the eye]

Table 2. Surface characteristics of different orifice materials used in the present experiments

Orifice material	3-phase contact angle ($^\circ$) (orifice, water and air)	Average roughness (μm)
Stainless-steel	80 ± 2	1.4 ± 0.4
Teflon	94.5 ± 4.5	7.7 ± 1.5
Acrylic sheet	70.8 ± 2	0.3 ± 0.07

When a third bubble coalesces with the bubble that resulted from the period-2 bubbling with coalescence at the orifice as described above, it is known as triple coalescence (referred to as period-3 bubbling here). Three distinct bubble formation periods can be clearly identified and they repeat regularly. In some of the experiments carried out in the present work with liquid of low surface tension ($\sigma = 0.037$ N/m, obtained by adding 12% 2-propanol in water), period-3 bubbling was observed as shown in Figure 4d.

In one of the literature reports (Tufail and Sartorelli, 2000), a period-4 bubbling regime that occurs due to coalescence of the third and fourth bubbles with a bubble cluster resulting from period-2 bubbling with coalescence at the orifice was also reported. In the period-4 bubbling regime, four distinct bubble formation periods that repeat regularly can be observed. In the present work (for higher gas flow rates in case of a 2 mm stainless-steel orifice), period-4 bubbling as suggested by Tufail and Sartorelli (2000), for a viscous system, could be observed (see Figure 2 in Tufail and Sartorelli, 2000). However, the scatter in the observed periods in the present study was much larger to identify clearly the period-4 bubbling regime. In a low-viscosity liquid (water) as used in the present work, the local velocity fluctuations created by bubble detachment or bubble-bubble coalescence above the orifice may be high enough to cause the observed scatter in bubble formation periods. In a

viscous system (glycerine + water, used in Tufail and Sartorelli, 2000, 2002), the local velocity fluctuations are damped, which minimizes the scatter in bubble formation periods (see Figure 3 in Tufail and Sartorelli, 2002).

After the period-3 or period-4 bubbling regime, the chaotic bubbling regime can be observed with further increase in the gas flow rate. The different bubble formation regimes observed in the present study agree well with the earlier investigations (McCann and Prince, 1971; Kyriakides et al., 1997; Tufail and Sartorelli, 2000; Zhang and Shoji, 2001; Tufail and Sartorelli, 2002). The objective of the present work was to understand how the orifice diameter, surface wetting characteristics, and surface tension of the liquid phase influence these bubbling regimes and their transition. The results on the effects of different parameters on bubbling regimes are discussed in the following sections.

Effect of Surface Wetting Characteristics

The effect of surface wetting characteristics was studied by considering bubble formation on stainless-steel ($\theta = 80^\circ$), Teflon ($\theta = 94.5^\circ$) and acrylic sheet ($\theta = 70.8^\circ$) orifice plates. The bubble formation periods, their average values of the individual periods (with lines) at different gas flow rates and for orifice diameters of 2, 4 and 6 mm are shown in Figures 5, 6 and 7, respectively. For a 2 mm diameter stainless-steel orifice, it can be seen from Figure 5a that the bubble formation belongs to period-1 bubbling up to a gas flow rate of about $330 \text{ cm}^3/\text{min}$. With further increase in gas flow rate, period-2 bubbling was observed, hence, there are two different bubble volumes exists at the same Q_G (Figures 5a and 5b). Since the transition between the bubble formation regimes occurs gradually (rather than an abrupt transition) and the measurement error in the bubble formation period is of the order of ± 1 ms, it is difficult to specify the exact value of gas flow rate at which the regime transition has occurred. As the gas flow rate increased, the transition from a period-2 bubbling regime to a chaotic bubbling regime was observed at a gas flow rate of about $943 \text{ cm}^3/\text{min}$.

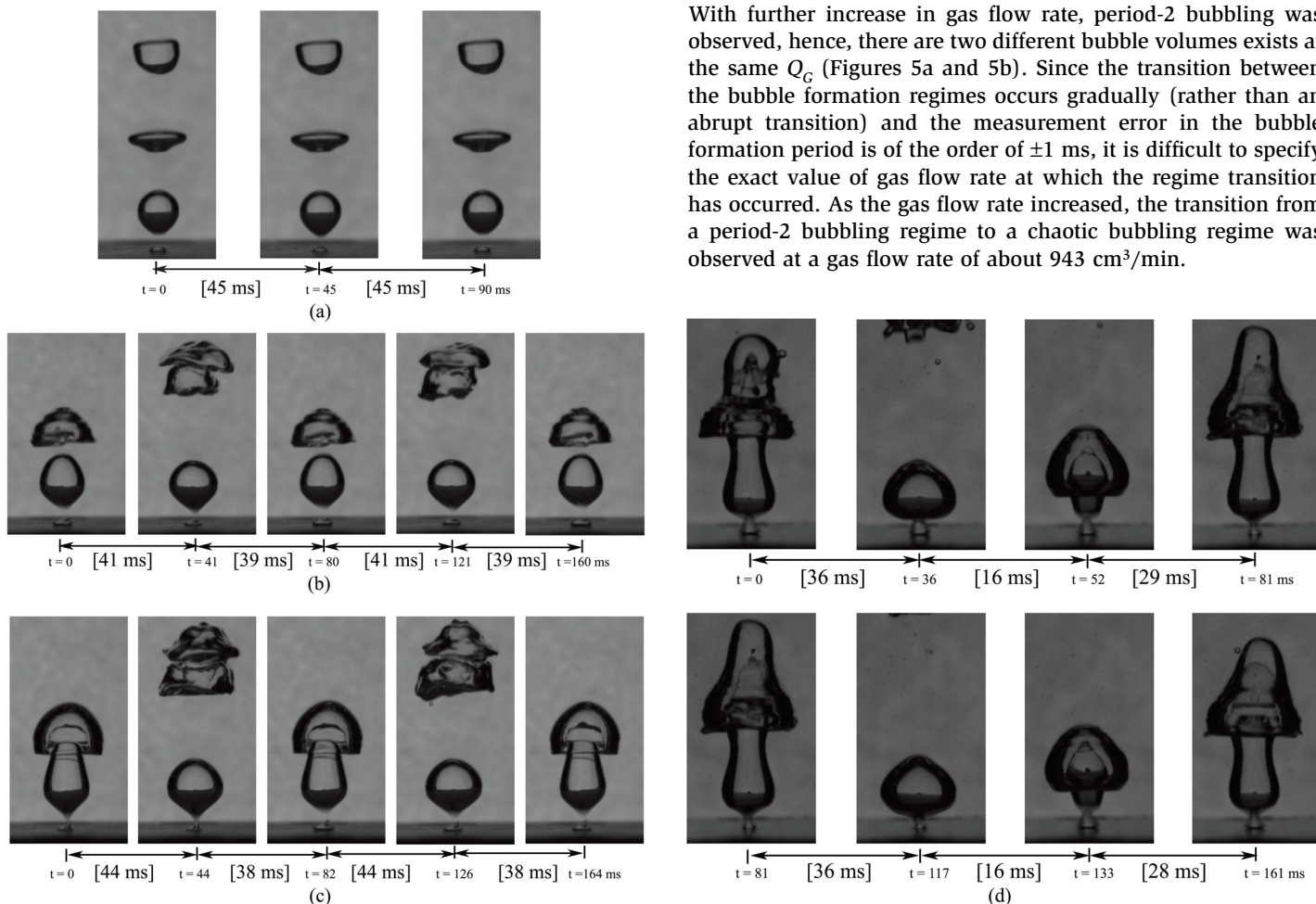


Figure 4. Regimes of bubble formation: (a) period-1 bubbling ($d_o = 2$ mm, Teflon orifice, $Q_G = 94 \text{ cm}^3/\text{min}$, liquid:water); (b) period-2 bubbling with pairing ($d_o = 2$ mm, Teflon orifice, $Q_G = 377 \text{ cm}^3/\text{min}$, liquid:water); (c) period-2 bubbling with coalescence ($d_o = 2$ mm, Teflon orifice, $Q_G = 566 \text{ cm}^3/\text{min}$, liquid:water); and (d) period-3 bubbling ($d_o = 2$ mm, stainless-steel orifice, $Q_G = 566 \text{ cm}^3/\text{min}$, liquid:water+12% 2-propanol)

It should be noted that the bubble formation period decreases (and the corresponding bubble formation frequency increases) with increase in gas flow rate until a point at which period-1 bubbling changes to period-2 bubbling. In this regime, the bubble formation is governed by the orifice diameter, surface wetting characteristics and liquid properties, apart from the gas flow rate. It can also be seen from Figure 5a that as the contact angle increased, the bubble formation period increased, for example, 40 and 45 ms for stainless-steel ($\theta = 80^\circ$) and Teflon ($\theta = 94.5^\circ$) orifice, respectively, for $d_o = 2$ mm and $Q_G = 94.5$ cm³/min. With increase in contact angle, the bubble spreads more on the orifice plate and eventually takes larger time for its detachment as indicated by the increase in the bubble formation periods. This agrees well with previous report in the literature (Gnyloskurenko et al., 2003). However, in spite of having a smaller contact angle ($\theta = 70.8^\circ$) for an acrylic orifice, the bubble formation period was between those with stainless-steel and Teflon orifices, for example, 42 ms for $d_o = 2$ mm and $Q_G = 94.5$ cm³/min, as seen from Figure 5a. This indicates that the effect of surface roughness also needs to be considered in

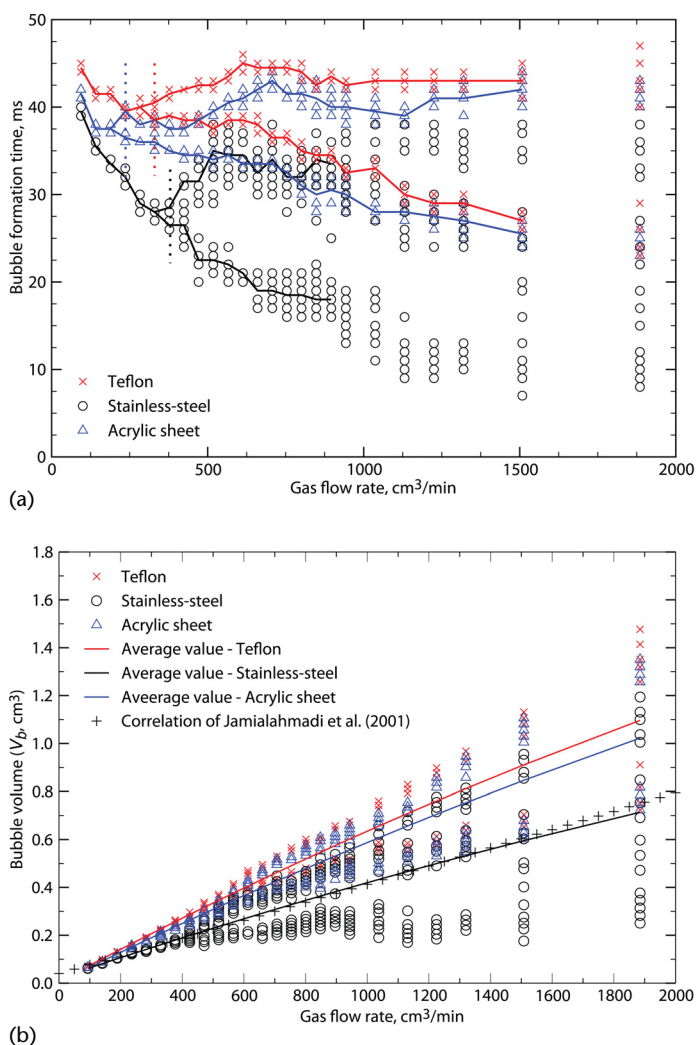


Figure 5. Effect of surface characteristics on: (a) bubble formation periods [lines indicate the average of the individual bubble formation periods corresponding to one branch for period-1 regime and two branches for period-2 regime. The dotted vertical lines denote the gas flow rate at which period-1 changes to period-2 bubbling]; and (b) bubble volume [averaged bubble volumes indicated with lines] ($d_o = 2$ mm, liquid:water)

addition to the effect of contact angle to understand the variation in bubble formation period. To empathize these effects further, the surface roughness was measured for the three orifice plates used in the present work (see Table 2). It was observed that although the acrylic orifice plate had a low contact angle, its average surface roughness ($0.3 \mu\text{m}$) was smaller than that of the stainless-steel (average roughness = $1.4 \mu\text{m}$) and Teflon orifice (average roughness = $7.7 \mu\text{m}$) plates. Therefore, the gas bubble was found to spread on the orifice plate, thus leading to a large bubble formation period than with a stainless-steel orifice.

To illustrate the effect of contact angle and surface roughness further, a sequence of bubble formations on different orifice materials is shown in Figure 8 for a $d_o = 2$ mm orifice and a gas flow rate of 377 cm³/min. It can be seen from Figure 8a that for a stainless-steel orifice ($\theta = 80^\circ$, average roughness = $1.4 \mu\text{m}$), the bubble base does not spread out of the orifice rim. However, for a Teflon orifice ($\theta = 94.5^\circ$, average roughness = $7.7 \mu\text{m}$), the bubble spreads significantly above the surface in spite of the higher roughness value (maximum spreading shown in Figure 8c(ii)). This suggests that the effect of contact angle was predominant compared with that of the roughness on the bubble formation period in the case of a Teflon orifice. For an acrylic orifice ($\theta = 70.8^\circ$, average roughness = $0.3 \mu\text{m}$), in spite of having a contact angle less than that of the stainless-steel orifice, the bubble base spreads outside the orifice rim (maximum spreading shown in Figure 8b(ii)) leading to an increased bubble formation period. This can be attributed due to the very small surface roughness of the acrylic sheet in comparison with the stainless-steel plate. In all cases, it was observed that the bubble base contracts again to the orifice rim before detachment occurs (image (iii) in Figures 8a, 8b and 8c). For more quantitative information on the effect of surface roughness on bubble formation behaviour, experiments in which the surface roughness is carefully varied are required. As the diameter of the orifice increased (Figures 6 and 7), the effect of surface characteristics on the bubble formation time in the period-1 bubbling regime decreased and almost vanished for the case of a 6 mm orifice (see period-1 bubbling regime in Figure 7). This can be clearly seen from the lines of average bubble formation periods.

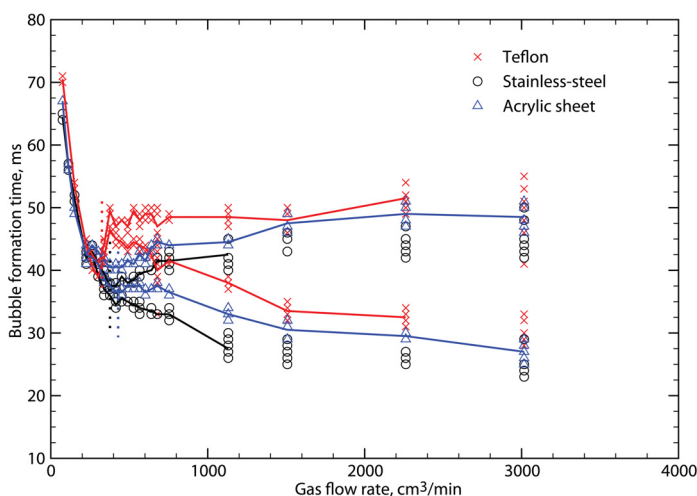


Figure 6. Effect of surface characteristics on bubble formation periods [lines indicate the average of the individual bubble formation periods corresponding to one branch for period-1 regime and two branches for period-2 regime. The dotted vertical lines denote the gas flow rate at which period-1 changes to period-2 bubbling] ($d_o = 4$ mm, liquid:water)

In the case of the period-2 bubbling regime, as indicated by Figures 5, 6 and 7, the bubble formation periods are always larger for Teflon and acrylic orifices than for a stainless-steel orifice. It should also be noted that the effect of surface wetting characteristics diminishes with increase in the orifice diameter. Further, one can observe from the Figures 5, 6 and 7 that the average bubble formation periods (or the average bubble formation frequency) remains constant in the period-2 bubbling regime, which corresponds to the constant frequency regime of Davidson and Schueler (1960b). However, considerable differences exist in bubble formation periods about the average value owing to aperiodic bubbling. Such a scatter of bubble formation periods about its average value at a particular Q_G has been completely ignored in the previous literature. The bubble volumes computed from the bubble formation periods reported for different gas flow rates are shown in Figure 5b. The corresponding average bubble volume is also shown with lines in the same figure. It can be seen that the average bubble volume increases with the gas flow rate, which agrees with the literature reports. It can also be seen that the maximum bubble volume observed at any particular gas flow rate was about 4–5 times greater than the minimum bubble volume, especially for higher flow rates (Figure 5b). Apparently, in most of the experimental data in the literature, only the average bubble volume was reported and several models (some of the models are given in Table 1) were developed to predict it. The model of Jamialahamadi et al. (2001) outperform the previous models reported in the literature. Therefore, the model of Jamialahamadi et al. (2001) (see Table 1) was considered in the present range of gas flow rate and the predictions were plotted in Figure 5b. It should be noted that although such models can predict the average bubble volume satisfactorily, these models have been derived based on the assumption of single bubbling. In period-2 and subsequent bubbling regimes, the observed bubble volumes vary significantly from the average value and the available models can not predict the bubble volumes observed in period-2 or period-3 bubbling regimes. Though the predictions of Jamialahamadi et al. (2001) agrees well with the bubble formation on the stainless-steel orifice, it should be noted that none of the present models can account for the effect of surface wetting characteristics

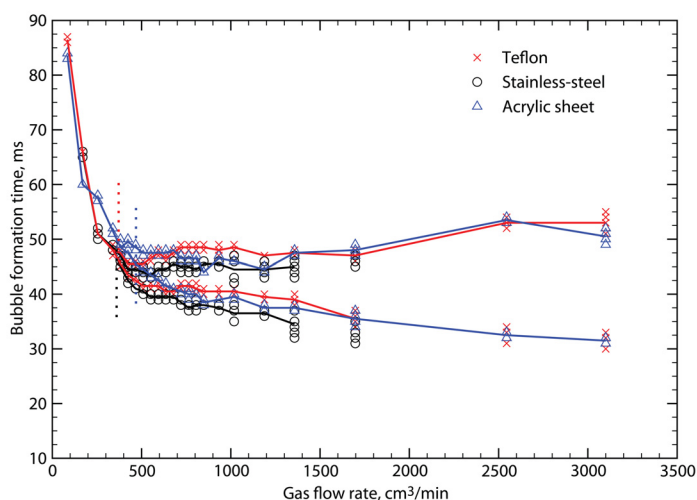


Figure 7. Effect of surface characteristics on bubble formation periods [lines indicate the average of the individual bubble formation periods corresponding to one branch for period-1 regime and two branches for period-2 regime. The dotted vertical lines denote the gas flow rate at which period-1 changes to period-2 bubbling] ($d_o = 6$ mm, liquid:water)

(Figure 5b). To develop non-linear models (Zhang and Shoji, 2001) or CFD-based models (Buwa et al., 2005; Gerlach et al., 2006) that can predict the aperiodic bubbling behaviour, the data presented in this paper can be extremely useful.

As seen from Figure 5a, it should be noted, however, that the transition from period-1 to period-2 bubbling occurs at smaller flow rates for Teflon and acrylic orifices than for a stainless-steel orifice. The onset of the period-2 bubbling regime and the corresponding critical gas flow rates are also marked in Figures 5, 6 and 7. The critical gas flow rates (or critical gas velocities) for orifices with different surface characteristics are given in Table 3. While chaotic bubbling was observed for a 2 mm stainless-steel orifice at a gas flow rate of 943 cm³/min, the bubble formation was still in the period-2 bubbling regime for Teflon and acrylic orifice plates. The experimental data on the critical gas flow rates for different regime transitions will be useful to identify the bubble formation regimes *a priori* from knowledge of the operating and design parameters.

Effect of Orifice Diameter

The effect of orifice diameter on bubble formation periods is shown in Figures 5, 6 and 7. It can be seen that the bubble formation period increases with increase in the orifice diameter at any particular gas flow rate. In particular, in the case of period-1 bubbling, the bubble formation period increases

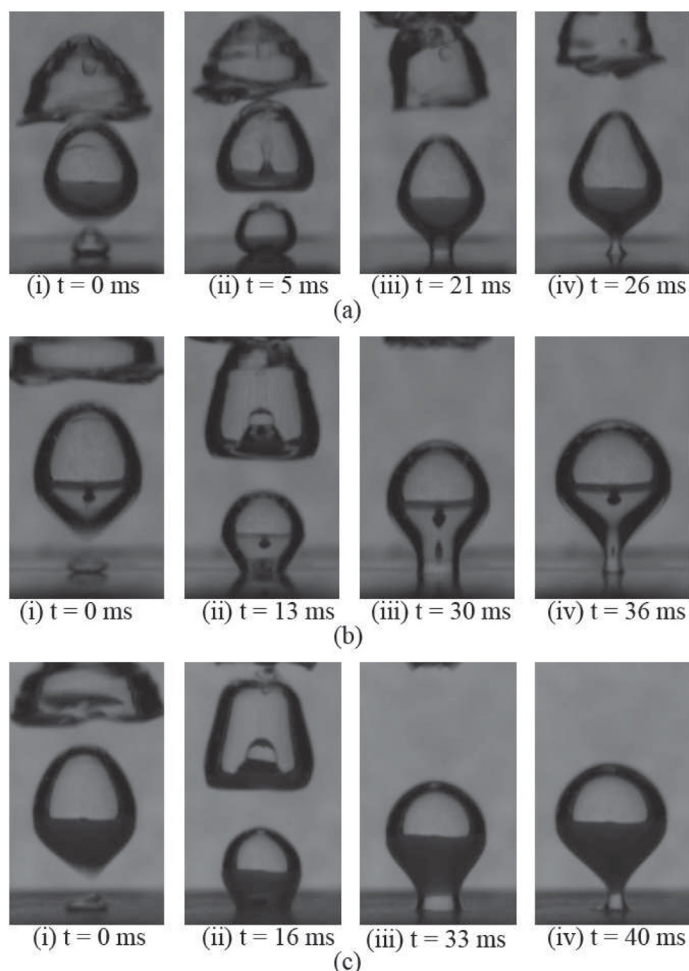


Figure 8. Effect of contact angle and roughness characteristics on spreading of the bubble during its formation on orifice plates of (a) stainless-steel; (b) acrylic sheet; and (c) Teflon ($d_o = 2$ mm, $Q_G = 377$ cm³/min, liquid:water)

Table 3. Critical gas flow rate (or gas velocity) at which bubble formation regime transition occurs from period-1 to period-2 and period-2 to chaotic bubbling for different orifice diameters and orifice materials

d_o (mm)	$Q_G (U_{oc})$ [cm^3/min (m/s)]					
	Stainless-steel		Teflon		Acrylic sheet	
	Period-1 to period-2	Period-2 to chaotic	Period-1 to period-2	Period-2 to chaotic	Period-1 to period-2	Period-2 to chaotic
2	377 (2.0)	943 (5.0)	330 (1.75)	1885 (10.0)	236 (1.25)	1885 (10.0)
4	302 (0.4)	1508 (2.0)	264 (0.35)	3016 (4.0)	264 (0.35)	–
6	382 (0.23)	1697 (1.0)	382 (0.23)	–	424 (0.25)	–

significantly with increase in orifice diameter. For example, at a gas flow rate of about $100 \text{ cm}^3/\text{min}$, bubble formation periods were around 40, 60 and 75 ms for 2, 4 and 6 mm diameter stainless-steel orifices, respectively (Figures 5, 6 and 7). At higher gas flow rates, i.e., in period-2 bubbling regime and chaotic bubbling regime, the effect of orifice diameter on bubble formation periods is not significant. It can be noted from the Figures 5, 6 and 7 that with increase in orifice diameter, the effect of surface wetting characteristics was diminished.

However, as seen from Figures 5, 6 and 7, the bubbling behaviour and its transition were found to be sensitive to the orifice diameter. For example, the critical gas velocity at which transition from period-1 to period-2 bubbling regime occurs decreases with increase in the orifice diameter. Table 3 gives quantitative information on how the critical gas velocity changes with change in orifice diameter for different materials. The critical velocity at which transition from period-2 to chaotic bubbling occurs also decreases with increase in diameter of the orifice. It should also be noted for Teflon and acrylic orifices that the chaotic bubbling regime was already established at a gas flow rate of $1885 \text{ cm}^3/\text{min}$ for a 2 mm orifice whereas for 4 and 6 mm orifices, the bubble formation process was still in the period-2 bubbling regime (see Table 3).

Effect of Surface Tension

The effect of surface tension on regimes of bubble formation was studied on a stainless-steel orifice plate ($d_o = 2 \text{ mm}$) by adding different amounts of 2-propanol to water ($\sigma = 0.052 \text{ N/m}$ with 4% 2-propanol in water and $\sigma = 0.037 \text{ N/m}$ with 12% 2-propanol in water (Satyanarayan et al., 1969)), and the measured bubble formation periods are shown in Figure 9. For the period-1 bubbling regime, it can be clearly seen from Figure 9 that the bubble formation period (and therefore bubble volume) decreases with decrease in surface tension (up to $Q_G = 188 \text{ cm}^3/\text{min}$). This agrees well with the earlier reports (Davidson and Schuele, 1960b; Ramakrishnan et al., 1969; Satyanarayan et al., 1969; Terasaka and Tsuge, 1993). After regime transition from period-1 to period-2 bubbling, no clear trend of the effect of surface tension on the bubble formation period can be seen for higher gas flow rates.

Interestingly, the critical gas velocity (or flow rate), at which regime transition from period-1 to period-2 bubbling occurred was found to decrease with decrease in surface tension. For example, for $\sigma = 0.072 \text{ N/m}$, the regime transition from period-1 to period-2 bubbling occurred at $U_{oc} = 2 \text{ m/s}$ ($Q_G = 377 \text{ cm}^3/\text{min}$), where as for $\sigma = 0.052 \text{ N/m}$, it occurred at $U_{oc} = 1.5 \text{ m/s}$ ($Q_G = 283 \text{ cm}^3/\text{min}$) and for $\sigma = 0.037 \text{ N/m}$, it occurred at $U_{oc} = 1 \text{ m/s}$ ($Q_G = 188 \text{ cm}^3/\text{min}$). It should also be noted that for the cases of $\sigma = 0.072$ and 0.052 N/m , the transition occurred directly from period-2 to the chaotic bubbling regime with an

increase in the gas flow rate. However, for $\sigma = 0.037 \text{ N/m}$, the period-3 bubbling regime, as shown in Figure 4d, was observed before the chaotic bubbling regime. This is discussed further in the following section.

Bubble Formation Regime Maps

The transition between various bubbling regimes can be represented by bubble formation regime maps constructed using appropriate dimensionless numbers. Kyriakides et al. (1997) presented the regimes of bubble formation with the help of the Bond number (Bo) and Froude number (Fr). The results discussed in the previous sections on the effects of various parameters on bubbling regimes are shown in Figures 10, 11 and 12 using Fr and Bo for bubble formation on orifice plates made of stainless-steel, Teflon and acrylic sheet, respectively. In general, it can be seen that for low Fr and Bo , the period-1 bubbling regime was observed. At any particular Fr , the regime transition from period-1 to period-2 bubbling occurs with increase in Bo (due to an increase in d_o or a decrease in σ in the present work). With increase in Fr (due to increase in U_o or decrease in d_o , in the present work), it was observed that bubbling regime transition occurs from period-1 to period-2 (with pairing and with coalescence at orifice) and finally to chaotic bubbling at a particular Bo . However, for $d_o = 2 \text{ mm}$ and $\sigma = 0.037 \text{ N/m}$, a distinct period-3 bubbling regime was also observed. It can be seen from Figure 10 that the range of the period-2 bubbling regime (in particular, period-2 bubbling

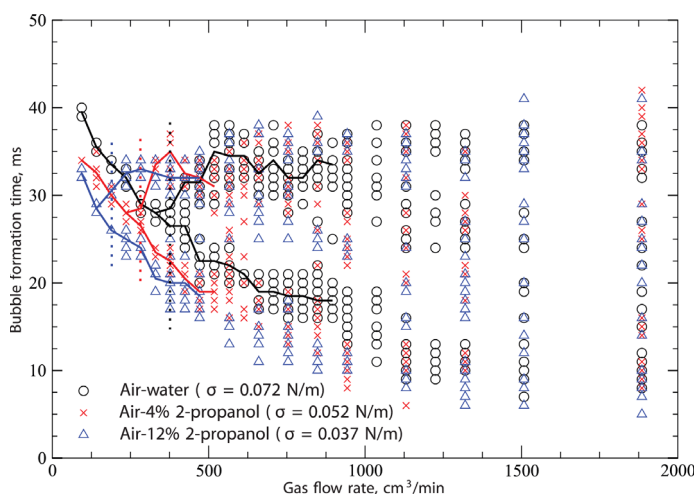


Figure 9. Effect of surface tension on bubble formation periods [lines indicate the average of the individual bubble formation periods corresponding to one branch for period-1 regime and two branches for period-2 regime. The dotted vertical lines denote the gas flow rate at which period-1 changes to period-2 bubbling] ($d_o = 2 \text{ mm}$, stainless-steel orifice).

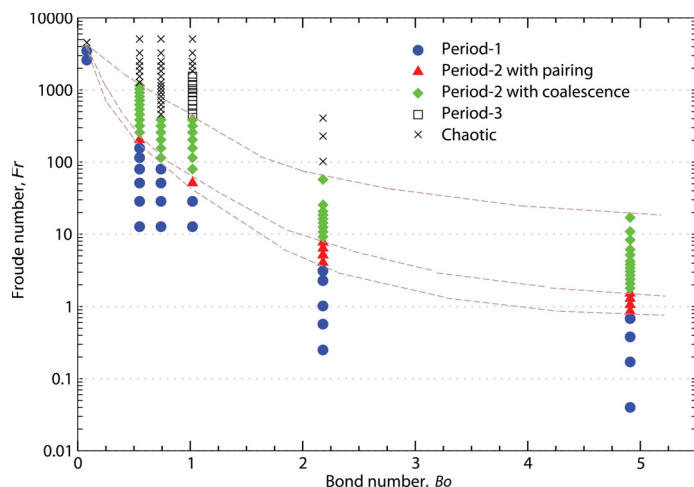


Figure 10. Bubble formation regime map for a stainless-steel orifice (liquid: water, water+2-propanol system) [The dotted lines are drawn to provide a guide to the eye]

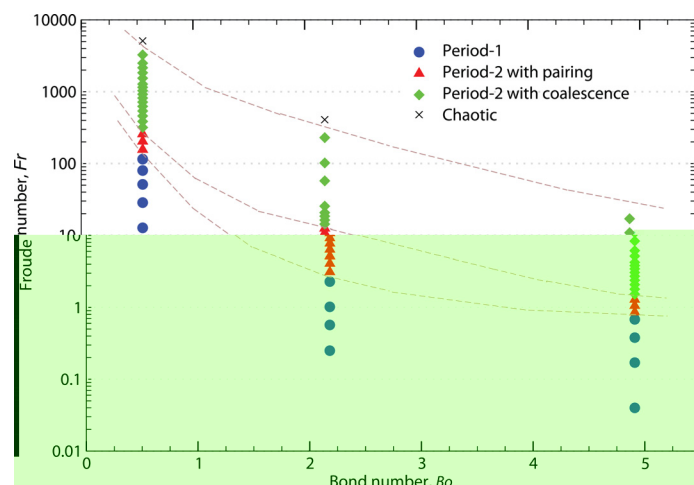


Figure 11. Bubble formation regime map for a Teflon orifice (liquid: water) [The dotted lines are drawn to provide a guide to the eye]

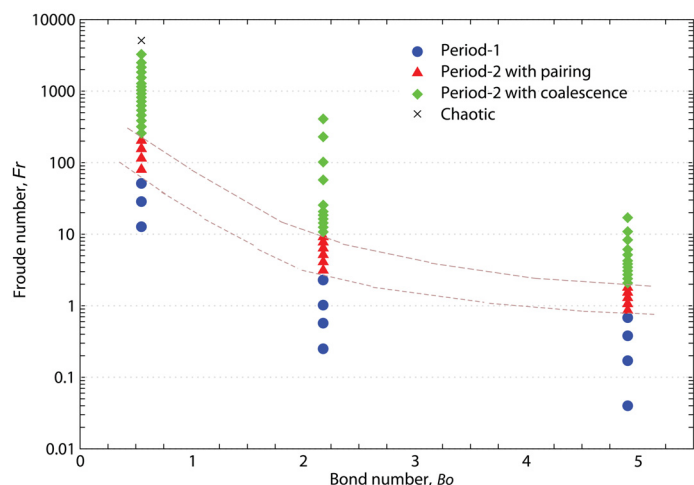


Figure 12. Bubble formation regime map for an acrylic sheet orifice (liquid: water) [The dotted lines are drawn to provide a guide to the eye]

regime with pairing) decreases with decrease in Bo . For a 2 mm stainless-steel orifice, a period-2 bubbling regime with pairing was observed only once in comparison with the significantly observed period-2 bubbling with pairing for 4 and 6 mm

stainless-steel orifices. Further, no period-2 bubbling regime (with pairing or with coalescence at the orifice) was observed (see Figure 10) in the literature (Tufaile and Sartorelli, 2002) reported for a smaller orifice diameter ($d_o = 0.78$ mm, $Bo = 0.083$). This suggests that for very small values of Bo , the period-1 bubbling regime directly changes to the chaotic bubbling regime.

For Teflon and acrylic orifices, the features of transition from period-1 to period-2 bubbling were not significantly affected by surface characteristics of the orifice material (see Figures 11 and 12). However, the period-2 bubbling regime (with pairing or with coalescence) and the transition between period-2 and chaotic bubbling was significantly affected by the surface characteristics (contact angle and surface roughness). It is required to perform the experiments of bubble formation by carefully varying the surface roughness of the orifice plates. With this, a generalized bubbling regime map can be provided using appropriately modified definitions of Bo and Fr that include the roughness and contact angle parameters. Efforts are under way to carry such experiments and to modify the dimensionless numbers to account for the effect of contact angle and surface roughness characteristics to provide generalized regime maps. The regime maps provided here can be used to identify the bubbling regimes *a priori* from the knowledge of the system parameters.

CONCLUSIONS

In this work, the different regimes of bubble formation were investigated experimentally under constant gas inflow conditions through a stagnant liquid. The effects of gas flow rate, orifice diameter, surface characteristics and surface tension on bubble formation periods and their regime transitions were studied. The following conclusions can be drawn:

- Depending on the magnitude of the gas flow rate, different bubbling behaviour, namely period-1, period-2 (with pairing or with coalescence at the orifice) and chaotic bubbling, were observed. For lower surface tension ($\sigma = 0.037$ N/m), a distinct period-3 bubbling regime was also observed. Therefore, the regime transition from period-1 to chaotic bubbling (or *the route to chaos*) depends not only on the gas flow rate, but also on the orifice diameter, surface characteristics and surface tension.
- The bubble volume or the bubble formation time increased with increase in diameter of the orifice. The critical orifice gas velocity or flow rate (at which the regime transition from period-1 to period-2 bubbling and from period-2 to chaotic bubbling takes place) decreased with increase in the diameter of the orifice. The observed range of the period-2 bubbling regime decreased as the orifice diameter was decreased. This agrees well with the literature report that for very small diameter orifices (for example, $d_o = 0.78$ mm (Tufaile and Sartorelli, 2002)), the transition occurred directly from the period-1 to the chaotic bubbling regime without the presence of a period-2 bubbling regime (Tufaile and Sartorelli, 2002).
- The effect of surface characteristics on the bubble volume or formation time were more significant for small orifices and the effect vanished as the orifice diameter increased. As the contact angle increased, it was observed rather qualitatively that the critical gas velocity at which transition from period-1 to period-2 bubbling occurs decreases. On smooth surfaces (low roughness value), the bubble spreads on the orifice plate leading to a large bubble formation period and, therefore, the

transition to period-2 bubbling was observed to occur earlier. However, the critical gas velocity at which transition from period-2 to chaotic bubbling occurs increases with increase in the contact angle.

- As the surface tension decreases, the bubble formation time or volume decreased in period-1 bubbling where as no significant influence was observed in period-2 bubbling regime. With decrease in surface tension, it was clearly observed that the critical gas velocity (or flow rate) at which the regime transition from period-1 to period-2 and from period-2 to chaotic bubbling occurs, decreased.

The physical understanding and the experimental data on the effects of various system parameters on different bubbling regimes reported in the present work are expected to contribute significantly for the further development and validation of analytical bubble formation models (Zhang and Shoji, 2001). It should be noted that the present analytical models (see Figure 1) cannot predict the aperiodic bubbling regimes and assumes to be single period bubbling at all flow rates. Present models cannot account for the effect of surface characteristics even on the prediction of average bubble volume (see Figure 5(b)). The experimental data presented in this work can be used to improve the models that were developed on the assumption of single bubbling. Such data can also be very crucial to validate CFD models, based on the VOF or LS method (Gerlach et al., 2006; 2007). In addition to the development of analytical or CFD models, the bubbling regime maps presented in this work can be very helpful for understanding bubbling behaviour in various gas-liquid contacting devices.

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NOMENCLATURE

Bo	Bond number ($= \rho_l g d_o^2 / \sigma$)
d	diameter, m
d_{avg}	average bubble diameter, m
d_{cap}	diameter of capillary, m
Fr	Froude number ($= U_o^2 / g d_o$)
g	gravitational acceleration, m/s ²
l_{cap}	length of capillary, m
P	pressure, N/m ²
Q_G	gas flow rate, m ³ /s
t	time, s
U	velocity, m/s
V	volume, m ³

Greek Symbols

ρ	density, kg/m ³
σ	surface tension, N/m
θ	contact angle, °
μ	viscosity, kg/m.s

Subscripts

b	bubble
c	chamber

l	liquid
g	gas
o	orifice
oc	critical

Acronyms

LS	level set
VOF	volume of fluid

REFERENCES

- Bork, O., M. Schlueter and N. Raebiger, "The Impact of Local Phenomena on Mass Transfer in Gas-Liquid Systems," *Can. J. Chem. Eng.* **83**(4), 658–666 (2005).
- Buwa, V., D. Gerlach and F. Durst, "Regimes of Bubble Formation on Submerged Orifices," *Phys. Rev. E*, submitted (2005).
- Cieslinski, J. and R. Mosdorf, "Gas Bubble Dynamics—Experiment and Fractal Analysis," *Int. J. Heat Mass Transfer* **48**, 1808–1818 (2005).
- Davidson, J. and B. Schueler, "Bubble Formation at an Orifice in a Viscous Liquid," *Trans. Inst. Chem. Eng.* **38**, 144–154 (1960a).
- Davidson, J. and B. Schueler, "Bubble Formation at an Orifice in an Inviscid Liquid," *Trans. Inst. Chem. Eng.* **38**, 335–342 (1960b).
- Davidson, J. F. and H. Harrison, "Fluidized Particles," Cambridge University Press, Cambridge, U.K. (1963).
- Davidson, L. and E. H. Amick, "Formation of Gas Bubbles at Horizontal Orifices," *AIChE J.* **2**(3), 337–342 (1956).
- Delnoij, E., J. Kuipers and W. van Swaaij, "Dynamic Simulation of Gas-Liquid Two-Phase Flow: Effect of Column Aspect Ratio on the Flow Structure," *Chem. Eng. Sci.* **52**(21), 3759–3772 (1997).
- Gaddis, E. S. and A. Vogelpohl, "Bubble Formation in Quiescent Liquids under Constant Flow Conditions," *Chem. Eng. Sci.* **41**(1), 97–105 (1986).
- Gerlach, D., N. Alleborn, V. Buwa and F. Durst, "Numerical Simulations of Periodic Bubble Formation at Submerged Orifices under Constant Inflow Conditions," *Chem. Eng. Sci.*, in press (2007).
- Gerlach, D., G. Biswas, F. Durst and V. Kolobaric, "Quasi-Static Bubble Formation on Submerged Orifices," *Int. J. Heat Mass Transfer* **48**, 425–438 (2005).
- Gerlach, D., G. Tomar, G. Biswas and F. Durst, "Comparison of Volume-of-Fluid Methods for Surface Tension-Dominant Two-Phase Flows," *Int. J. Heat Mass Transfer* **49**, 740–754 (2006).
- Gnyloskurenko, S., A. Byakova, O. Raychenko and T. Nakamura, "Influence of Wetting Conditions on Bubble Formation at Orifice in an Inviscid Liquid. Transformation of Bubble Shape and Size," *Colloids Surf. A: Physicochem. Eng. Aspects* **218**, 73–87 (2003).
- Harris, J. and H. Stoecker, "Handbook of Mathematics and Computational Science," Springer-Verlag, New York (1998).
- Jamialahmadi, M., M. R. Zehtaban, H. Mueller-Steinhagen, A. Sarrafi and J. M. Smith, "Study of Bubble Formation under Constant Flow Conditions," *Trans. Inst. Chem. Eng.* **79**, 523–532 (2001).
- Kulkarni, A. A. and J. B. Joshi, "Bubble Formation and Bubble Rise Velocity in Gas-Liquid Systems: A Review," *Ind. Eng. Chem. Res.* **44**, 5873–5931 (2005).

- Kumar, R. and N. Kuloor, "The Formation of Bubbles and Drops," *Adv. Chem. Eng.* **8**, 255–368 (1970).
- Kupferberg, A. and G. Jamson, "Bubble Formation at a Submerged Orifice Above a Gas Chamber of Finite Volume," *Trans. Inst. Chem. Eng.* **47**, 241–250 (1969).
- Kyriakides, N., E. Kastrinakis and S. Nychas, "Bubbling from Nozzles Submerged in Water: Transitions between Bubbling Regimes," *Can. J. Chem. Eng.* **75**, 684–691 (1997).
- Leibson, I., E. G. Holcomb, A. G. Cacosso and J. J. Jacmic, "Rate of Flow and Mechanics of Bubble Formation from Single Submerged Orifices," *AIChE J.* **2**, 296–306 (1956).
- Lin, T. J., K. Tsuchiya and L. S. Fan, "Bubble Flow Characteristics in Bubble Columns at Elevated Pressure and Temperature," *AIChE J.* **44**(3), 545–560 (1998).
- Liow, J.-L. and N. B. Gray, "A Model of Bubble Growth in Wetting and Non-Wetting Liquids," *Chem. Eng. Sci.* **43**(12), 3129–3139 (1988).
- McCann, D. and R. Prince, "Regimes of Bubbling at a Submerged Orifice," *Chem. Eng. Sci.* **26**, 1505–1512 (1971).
- Mittoni, L. J., M. P. Schwarz and R. D. L. Nauze, "Deterministic Chaos in the Gas Inlet Pressure of Gas-Liquid Bubbling Systems," *Phys. Fluids* **7**(4), 891–893 (1995).
- Mosdorf, R. and M. Shoji, "Chaos in Bubbling—Nonlinear Analysis and Modelling," *Chem. Eng. Sci.* **58**, 3837–3846 (2003).
- Nguyen, K., C. Daw, P. Chakka, M. Cheng, D. Bruns and C. Finney, "Spatio-Temporal Dynamics in a Train of Rising Bubbles," *Chem. Eng. J.* **65**, 191–197 (1996).
- Park, Y., A. L. Tyler and N. de Nevers, "The Chamber Orifice Interaction in the Formation of Bubbles," *Chem. Eng. Sci.* **32**, 907–916 (1977).
- Raebiger, N. and A. Vogelpohl, "Calculation of Bubble Size in Stationary and Moving Newtonian Liquids," *Chem. Ing. Tech.* **54**(11), 1082–1083 (1982).
- Ramakrishnan, S., R. Kumar and N. Kuloor, "Studies in Bubble Formation – I, Bubble Formation under Constant Flow Conditions," *Chem. Eng. Sci.* **24**, 731–747 (1969).
- Satyanarayan, A., R. Kumar and N. Kuloor, "Studies in Bubble Formation – II, Bubble Formation under Constant Pressure Conditions," *Chem. Eng. Sci.* **24**, 749–761 (1969).
- Takahashi, T. and T. Miyahara, "Bubble Volume Formed at Submerged Nozzles: Constant Gas Flow Condition," *Kogaku Kogaku Ronbunshu* **2**, 138–143 (1976).
- Terasaka, K. and H. Tsuge, "Bubble Formation at a Single Orifice in Highly Viscous Liquids," *J. Chem. Eng. Jpn.* **23**(2), 160–165 (1990).
- Terasaka, K. and H. Tsuge, "Bubble Formation under Constant-Flow Conditions," *Chem. Eng. Sci.* **48**(19), 3417–3422 (1993).
- Tritton, D. and C. Egde, "Chaotic Bubbling," *Phys. Fluids A* **5**(2), 503–505 (1993).
- Tsuge, H. and S.-I. Hibino, "Bubble Formation from an Orifice Submerged in Liquids," *Chem. Eng. Commun.* **22**, 63–79 (1983).
- Tufaile, A. and J. Sartorelli, "Chaotic Behaviour in Bubble Formation Dynamics," *Physica A* **275**, 336–346 (2000).
- Tufaile, A. and J. Sartorelli, "Bubble and Spherical Air Shell Formation Dynamics," *Phys. Rev. E* **66**, 056204-1-7 (2002).
- von Krevelen, D. and P. Hoftijzer, "Studies of Gas-Bubble Formation, Calculation of Interfacial Area in Bubble Contactors," *Chem. Eng. Prog.* **46**(1), 29–35 (1950).
- Walters, J. and J. Davidson, "The Initial Motion of a Gas Bubble Formed in an Inviscid Liquid," *J. Fluid Mech.* **17**, 321–336 (1963).
- Zhang, L. and M. Shoji, "A Periodic Bubble Formation from a Submerged Orifice," *Chem. Eng. Sci.* **56**, 5371–5381 (2001).

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